The Structure of the Elements

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Abstract

The Natural Philosophy of Universal Matter concludes with the creation of a hydrogen atom. How, then, are the remaining elements formed? This spreadsheet describes the construction of elements by placing protons and neutrons in specified sequences, accounting for all elements and their common isotopes. This document illustrates portions of the spreadsheet to enable interested readers to work through the structured build.

1 General Statements

Tabs In the build spreadsheet titled The-Structure-of-Elements.xlsx, Figure 1 shows the two tabs, "Itemized" for build detail and "Graphic" for a visual representation of the build detail, displayed side by side. If possible, it is best to view the two sheets side by side, as illustrated in Figure 2.

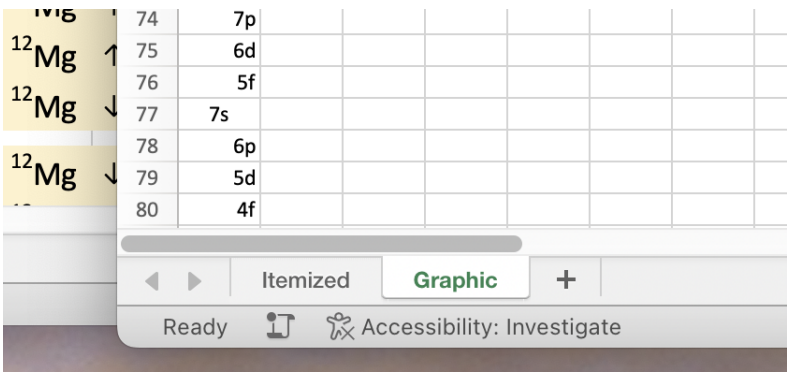


Figure 1: Showing Both Tabs

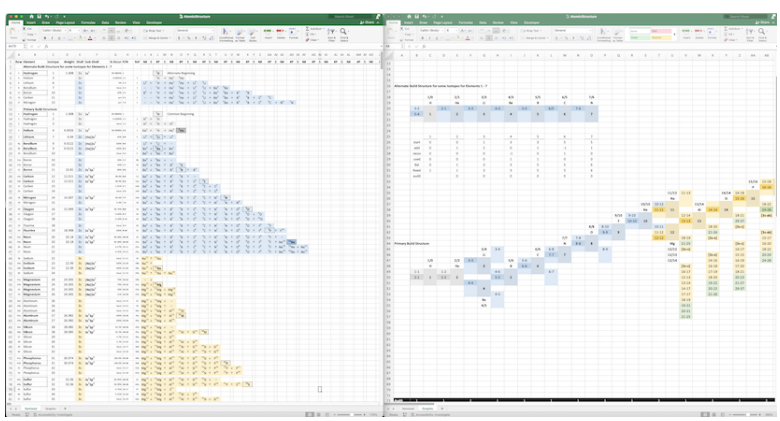


Figure 2: List and Graphic Side by Side

Frozen Panes Figure 3 shows the Itemized tab, with columns A though I set as a frozen pane:

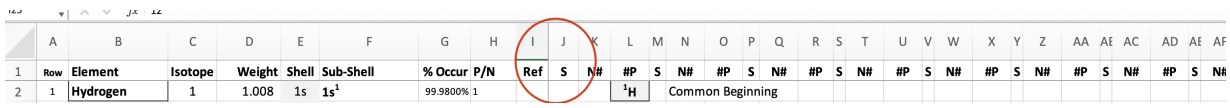


Figure 3: Frozen Panes

1.1 Color Coding

1.1.1 Columns and Rows

Both the Itemized and Graphic tabs illustrate an atom's shells in a color-coded format. For example, in Figure 4, the Itemized tab on the left and the Graphic tab on the right show that the first shell contains two elements, hydrogen and helium. Rows corresponding to this shell and these elements are color-coded gray. The second shell, containing lithium through neon, is color-coded medium blue in each tab. The color coding is applied consistently, even when transitioning between shells.

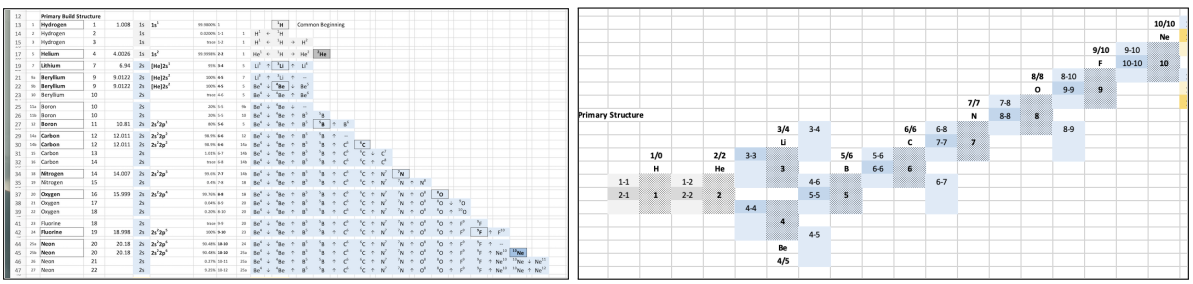


Figure 4: Color Coding

1.1.2 Final Elements in Shells

Each shell has a final element, such as helium in shell one and neon in shell two. These final elements are denoted by an emphasized color, for example darker gray for helium and darker blue for neon. See the arrows in Figure 5.

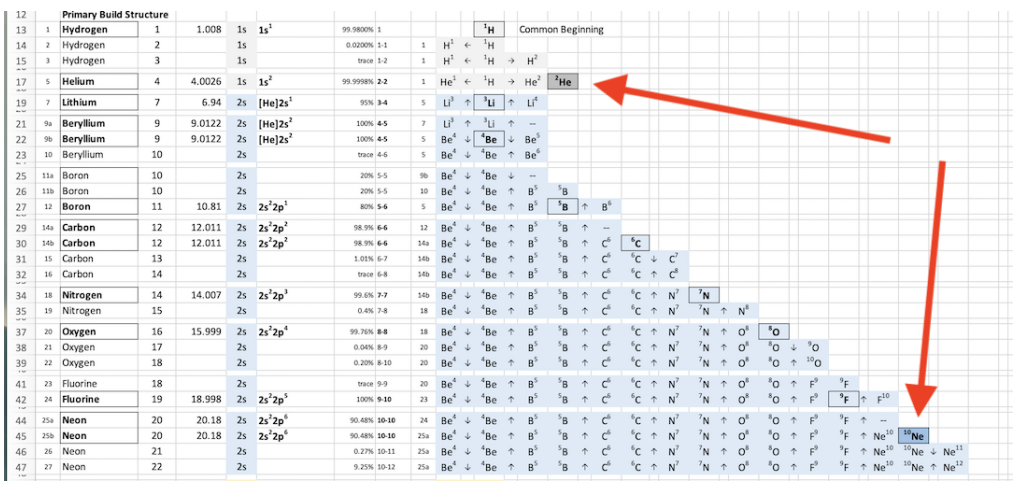


Figure 5: Final Shell Elements

2 Building Structures for Elements

2.1 Atom Rotation

The build structure, as noted on the Graphic tab, is written from left to right. However, the hydrogen proton rotates end over end, and as neutrons and then protons are added to the chain, their additional rotations are incorporated into the system. The rotations of the triple primes allow the chain to rotate and loop back on itself, forming spiral layers as the atoms increase in complexity.

The rotational complexity, even when adding the first neutron to hydrogen, demonstrates the difficulty of growing the build. Not only must neutrons and protons align with the neighboring triple prime, but they must also match its rotational velocity. The required pressure must be strong and persistent. Therefore, very few atoms are formed without additional assistance.[1] These include hydrogen, helium, and trace amounts of lithium and beryllium.

2.2 Primary (Common) and Alternate (Beginning) Structures

Figure 6 and Figure 7 detail two structures, one representing a standard build and the other an alternate build. The alternate structure is limited to the first few beginning elements, hydrogen through nitrogen, and some of their uncommon isotopes. The primary structure details the build for all remaining elements and their isotopes. See the arrows and text in the following figures.

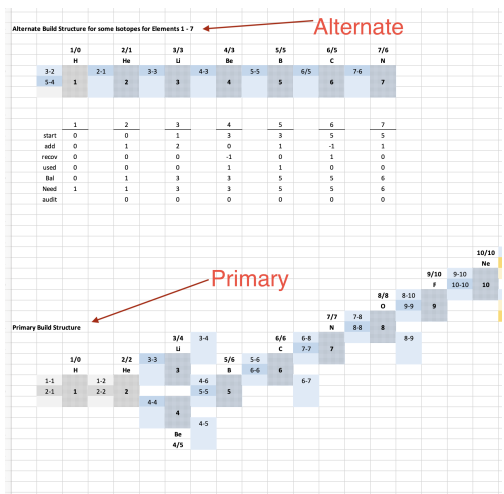


Figure 6: Graphic Build Structures

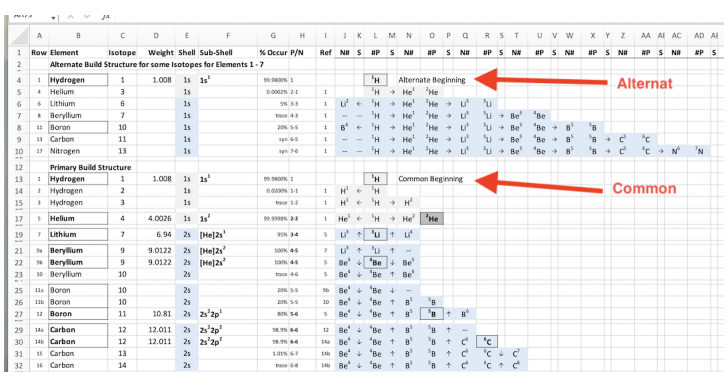


Figure 7: Itemized Element Structures

[1] Such as at the end of a star's lifetime or in a laboratory.

2.3 Understanding the Itemized Tab

2.3.1 Itemized Tab Columns

The first group of columns, A through I, of the Itemized tab, as shown in Figure 8, contains the following details:

Row: Numbers each isotope. If an isotope requires additional rows, a letter is added to the number. For example, beryllium, which requires two header rows, is marked 9a and 9b.

Element: Name of the element. A box is placed around primary isotopes for ease of reading. If an isotope spans multiple rows, as with beryllium, all of its rows are boxed together.

Isotope: Similar to Row but uses only a cardinal number.

Weight: Listed for primary isotopes only, for example header row 1 for hydrogen and header row 5 for helium.

Shell: Shell number, following common documentation such as Wikipedia.

Subshell: A more detailed shell designation. See Figure 16, box A, for details of this notation.

% Occur: Percentage of natural occurrence. Synthetic elements are marked "syn."

P/N: Number of protons and neutrons in the element.

Ref: A number referencing the previous row from which the build detail, beginning in column J, continues.

2.3.2 Shell Folding

Each color-coded shell is folded underneath the previous shell, as illustrated in Figure 8. Spreadsheet column A, with the header labeled "Row," marked "5," illustrates the last element of the first shell, helium. The first element of the second shell, lithium, is then moved to the header labeled "Row," marked "7." Lithium therefore starts in spreadsheet column J, just as hydrogen does in the first shell.

For readability, after the final element of any shell, the first and subsequent elements of the new shell are folded down to spreadsheet column J and assigned a new color.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AE	AC	AD	AE	A
1	Row	Element	Isotope	Weight	Shell	Sub-Shell	% Occur	P/N	Ref	N#	S	#P	S	N#	#P	S	N#	#P	S	N#	#P	S	N#	#P	S	N#	#P	S	N#	#P	S	N
2		Alternate Build Structure for some Isotopes for Elements 1 - 7																														
4	1	Hydrogen	1	1.008	1s	$1s^1$	99.9800%	1				^1H		Alternate Beginning																		
5	4	Helium	3		1s		0.0002%	2-1	1			^1H	→	He^1	^2He																	
6	6	Lithium	6		1s		5%	3-3	1	Li^2	←	^1H	→	He^1	^2He	→	Li^3	^3Li														
7	8	Beryllium	7		1s		trace	4-3	1	--	--	^1H	→	He^1	^2He	→	Li^3	^3Li	→	Be^3	^4Be											
8	11	Boron	10		1s		20%	5-5	1	B^4	←	^1H	→	He^1	^2He	→	Li^3	^3Li	→	Be^3	^4Be	→	B^5	^5B								
9	13	Carbon	11		1s		syn	6-5	1	--	--	^1H	→	He^1	^2He	→	Li^3	^3Li	→	Be^3	^4Be	→	B^5	^5B	→	C^5	^6C					
10	17	Nitrogen	13		1s		syn	7-6	1	--	--	^1H	→	He^1	^2He	→	Li^3	^3Li	→	Be^3	^4Be	→	B^5	^5B	→	C^5	^6C	→	N^6	^7N		
12		Primary Build Structure																														
13	1	Hydrogen	1	1.008	1s	$1s^1$	99.9800%	1				^1H		Common Beginning																		
14	2	Hydrogen	2		1s		0.0200%	1-1	1	H^1	←	^1H																				
15	3	Hydrogen	3		1s		trace	1-2	1	H^1	←	^1H	→	H^2																		
17	5	Helium	4	4.0026	1s	$1s^2$	99.9998%	2-2	1	He^1	←	^1H	→	He^2	^2He																	
19	7	Lithium	7	6.94	2s	$[He]2s^1$	95%	3-4	5	Li^3	↑	^3Li	↑	Li^4																		
21	9a	Beryllium	9	9.0122	2s	$[He]2s^2$	100%	4-5	7	Li^3	↑	^3Li	↑	--																		
22	9b	Beryllium	9	9.0122	2s	$[He]2s^2$	100%	4-5	5	Be^4	↓	^4Be	↓	Be^5																		
23	10	Beryllium	10		2s		trace	4-6	5	Be^4	↓	^4Be	↑	Be^6																		
25	11a	Boron	10		2s		20%	5-5	9b	Be^4	↓	^4Be	↓	--																		
26	11b	Boron	10		2s		20%	5-5	10	Be^4	↓	^4Be	↑	B^5	^5B																	
27	12	Boron	11	10.81	2s	$2s^22p^1$	80%	5-6	5	Be^4	↓	^4Be	↑	B^5	^5B	↑	B^6															

Figure 8: Itemized Column Header

2.3.3 Itemized Build Columns

Figure 9 illustrates a second header grouping consisting of a repeated three-column set labeled "S," "N#," and "#P," for example columns J, K, and L.

- S: Build subitem directional sign. From the last item or subitem, add the neutron or proton in the indicated direction: up, down, right, or left.
- N#: Element abbreviation with neutron number in superscript.
- #P: Proton number in superscript followed by the element abbreviation.

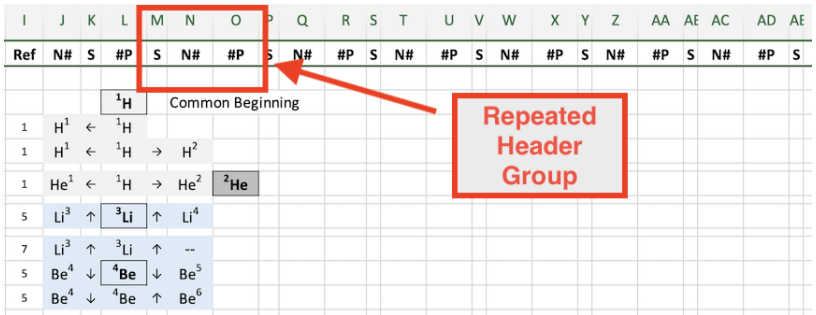


Figure 9: Itemized Build Column Header

2.4 Understanding the Graphic Tab

2.4.1 Graphic Chart Color Coding

As illustrated in Figure 10, color coding aids interpretation of the Graphic build chart. Items of interest include:

Label A: Proton marking, shown in the darker base color with a small dot fill.

Label B: Neutron marking, shown as a lighter variant of the shell's base color, indicating that it was used to designate the second neutron for hydrogen.

Label C: Neutron marking, shown in the darker base color, indicating that it was the final neutron used in conjunction with an element's proton and will remain in this position throughout the remainder of the chart. In the example, it denotes the second neutron used with helium.

Label D: At least two indicators appear above each proton. In this example, the first indicator is the abbreviation "B" for boron. The second indicator above or below the abbreviation is the proton-neutron number combination for a boxed element or isotope.

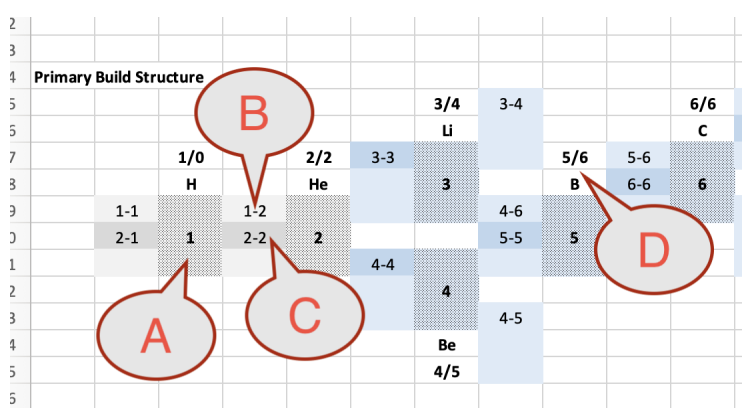


Figure 10: Graphic Color Coding

In some cases, multiple proton-neutron combinations may be listed in reverse ascending order, as illustrated in Figure 11. This occurs when an element has more than one primary isotope, such as ruthenium, element 44 on the periodic table, which has five naturally occurring isotopes each exceeding 10 percent prevalence.

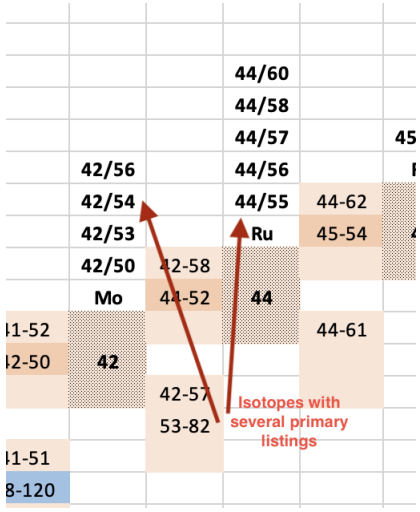


Figure 11: Graphic Showing Primary Isotopes

In some cases, the neutron list found between protons contains more than a limit of three neutrons. In these instances, a cross-referenced list is created above or below an element, as illustrated in Figure 12 for magnesium, element 12.

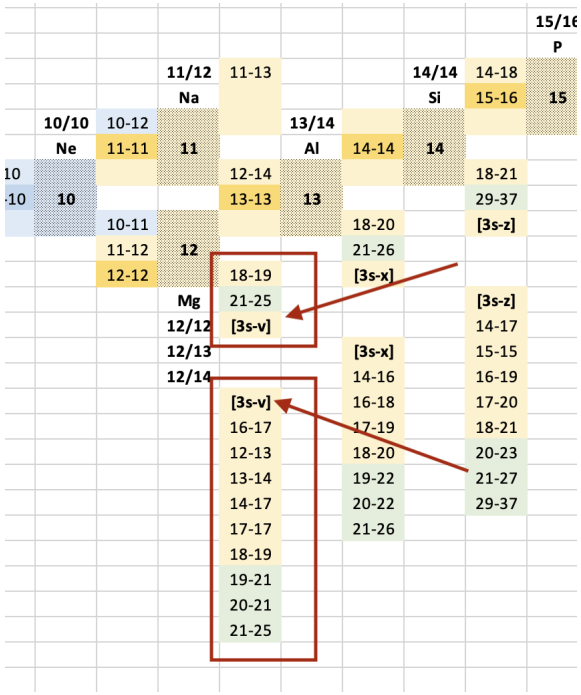


Figure 12: Extended Neutron Lists

The smaller rectangle in Figure 12 contains the highest cell labeled "18–19," representing the final neutron used to complete the third shell. The middle cell labeled "21–25" refers to the next-to-last neutron used in this position in the fourth shell. The bottom cell labeled "[3s-v]" refers to the extended list in the longer rectangle below. The bracketed reference includes the shell and subshell designation, followed by the Excel column letters indicating where the extended list is located.

3 Examples

3.1 Hydrogen and Helium

Building helium from hydrogen demonstrates the nomenclature of the Itemized and Graphic tabs. Hydrogen is detailed first. On the Itemized tab in Figure 13, header row 1 shows the boxed hydrogen entry and its single proton labeled $\boxed{^{1}\mathrm{H}}$. Two hydrogen isotopes are then added: one with one neutron in header row 2 and one with two neutrons in header row 3.

12		**Primary Build Structure**														
13	1	**Hydrogen**	1	1.008	1s	**$1s^1$**	99.9800%	1				**^{1}H**		Common Beginning		
14	2	Hydrogen	2		1s		0.0200%	1-1	1	H^1	←	^{1}H				
15	3	Hydrogen	3		1s		trace	1-2	1	H^1	←	^{1}H	→	H^2		
17	5	**Helium**	4	4.0026	1s	**$1s^2$**	99.9998%	**2-2**	1	He^1	←	^{1}H	→	He^2	**^{2}He**	

Figure 13: Sample Itemized Tab

44	**Primary Build Structure**			
45				
46				
47		**1/0**		**2/2**
48		**H**		**He**
49	1-1		1-2	
50	2-1	**1**	2-2	**2**
51				
52				
53				

Figure 14: Sample Graphic Tab

In Figure 13, hydrogen information begins in header row 1, with the hydrogen proton $\boxed{^{1}\mathrm{H}}$ located in column L and labeled "Common Beginning." Two neutrons are required to add the two hydrogen isotopes. The first added neutron is in header row 2, and the second is in header row 3. The red text and arrow labeled A indicate that the first neutron is added to the left of the proton labeled $\boxed{^{1}\mathrm{H}}$. The red text and arrow labeled B indicate that the second neutron is added to the right of the proton labeled $\boxed{^{1}\mathrm{H}}$.

The build instructions in Figure 13, header row 2, are repeated in header row 3, but new build instructions are added. The new build instructions place the second neutron to the right of hydrogen proton.

In Figure 14, the addition of the two neutrons on the Itemized tab is illustrated on the Graphic tab and marked "C" and "D." The first neutron is added to the left of the proton, labeled as proton-neutron pair "1-1," and indicated by the red arrow and text marked C. The second proton-neutron pair, "1-2," is added to the right of the proton and marked by the red arrow and text marked D.

Now the second proton, required for helium, is added. On the Itemized tab, in header row 5, the new proton is placed next to the second neutron of hydrogen, and labeled $\boxed{^{2}\mathrm{He}}$.

On the Graphic tab, the second proton is noted as "2" in the middle of the darker three-cell area. A bold text label "2-2" is placed in spreadsheet row 47 and "He" in the cell below in spreadsheet row 48. This completes the build chart for helium and completes the first atomic shell.[2]

[2]The special coloring for this cell is addressed in discussions around Figure 5.

On the Itemized tab, header row 5 contains the instructions to build helium. This build requires only a single instruction row. The neutrons for the first and second hydrogen isotopes are relabeled as neutrons for helium, that is proton-neutron pairs “2-1” and “2-2.” These neutron designations remain for the remainder of the build chart.

3.2 Lithium, Beryllium, and Boron

The first shell of the periodic table is complete. Next are the first three elements of the second shell.

17	5	**Helium**	4	4.0026	1s	**$1s^2$**		99.9998%	**2-2**	1	He^1	←	^1H	→	He^2	**^2He**	
19	7	**Lithium**	7	6.94	2s	**[He]$2s^1$**		95%	**3-4**	5	Li^3	↑	**^3Li**	↑	Li^4		
21	9a	**Beryllium**	9	9.0122	2s	**[He]$2s^2$**	A	100%	**4-5**	7	Li^3	↑	^3Li	↑	--		
22	9b	**Beryllium**	9	9.0122	2s	**[He]$2s^2$**		100%	**4-5**	5	Be^4	↓	**^4Be**	↓	Be^5		
23	10	Beryllium	10		2s			trace	4-6	5	Be^4	↓	^4Be	↑	Be^6		
25	11a	Boron	10		2s			20%	5-5	9b	Be^4	↓	^4Be	↓	--		
26	11b	Boron	10		2s			20%	5-5	10	Be^4	↓	^4Be	↑	B^5	^5B	
27	12	**Boron**	11	10.81	2s	**$2s^2 2p^1$**		80%	**5-6**	5	Be^4	↓	^4Be	↑	B^5	**^5B**	↑ B^6

Figure 15: Second Shell: Itemized Tab

43										
44	**Primary Build Structure**									
45						**3/4**	3-4			
46						**Li**				
47			**1/0**		**2/2**	3-3		**5/6**	5-6	
48			**H**		**He**		**3**	**B**	6-6	
49		1-1		1-2			4-6			
50		2-1	**1**	2-2	**2**		5-5	**5**		
51						4-4				
52							**4**			
53							4-5			
54						**Be**				
55						**4/5**				
56										
57										

Figure 16: Second Shell: Graphic Tab

Figure 15 illustrates the Itemized tab and identifies rows for lithium through boron, continuing after header row 5, helium. The red box labeled A on the Itemized tab lists the first two element entries of the new shell, lithium and beryllium. The shell details shown in the box are bracketed with “[He]” to indicate that the new shell continues after the final element of the previous shell, helium.

Also in Figure 15 in the red text box A, is the label on header rows 9a and 9b. This indicates that beryllium is the second proton in shell 2, subshell 2 and, on the Graphic tab in Figure 16, proceeds downward from helium.

The label “[He]$2s^1$” in header row 7 for lithium means it continues directly from helium and constitutes the first proton on shell 2, subshell s. However, in Figure 16 on the Graphic tab, lithium steps up from helium and constitutes the only element in shell 2, subshell 1.

Observe the proton-neutron nomenclature for lithium and beryllium on the Itemized tab. On header row 7, lithium contains three protons and four neutrons, recorded as 3-4 in spreadsheet column H. On rows 9a and 9b, beryllium contains four protons and five neutrons, recorded as 4-5 in the same column. This correlates to the Graphic tab for proton-neutron pairs 3-3 and 3-4 for lithium and 4-4 and 4-5 for beryllium.

Proton-neutron pair 3-3 is offset from helium and 3-4 is offset from proton 3, each positioned one cell horizontally and two cells upward. In the same manner, neutron 4-4 is offset from helium and 4-5 is offset from beryllium one cell horizontally and two cells downward. This offset is structural and found in nature. In these cases, the effective

and ineffective primes connect to the compatible ends of the other prime. Proton-neutron pair 3-3 connects squarely against the primes of the lithium proton. Pair 3-4 differs slightly, with the ineffective prime of the 3-4 neutron connecting to the effective prime of the lithium proton.

Note the difference in font for proton-neutron pairs 3-3 and 3-4 and 4-4 and 4-5. Pairs 3-3 and 4-4 are in bold text and a darker shade of blue, whereas 3-4 and 4-5 are not. The bold font indicates that these proton-neutron pairs are fixed in position and continue throughout the build. The pairs 3-4 and 4-5 are not permanent and disappear, with the positions remaining unused for other elements. These empty neutron positions are noted on the Itemized tab in row 9a for beryllium and row 11a for boron. The double-dash entries indicate that the underlying neutrons are deleted and subsequent build instructions do not replace them.

The beryllium isotope in spreadsheet row 49 adds a sixth neutron labeled 4-6 such that it straddles proton 3 for lithium and proton 4 for beryllium. The two outer ineffective primes of neutron 4-6 attach to the two outer effective primes of the lithium and beryllium protons. To build boron, this sixth neutron is renamed neutron 5-5, as noted in header row 11b on the Itemized tab. Boron's proton, the fifth added to the build, attaches to this neutron.

3.3 An H_2O (Water) Molecule

An H_2O molecule, water, is formed by adding two hydrogen atoms to oxygen. This is accomplished by adding a hydrogen atom to the top of lithium's neutron 3-3 and one to the bottom of beryllium's neutron 4-4. These are noted by the red text arrows in Figure 17. Note the circled join points showing hydrogen attaching to the outer ineffective primes of neutrons 3-3 and 4-4.

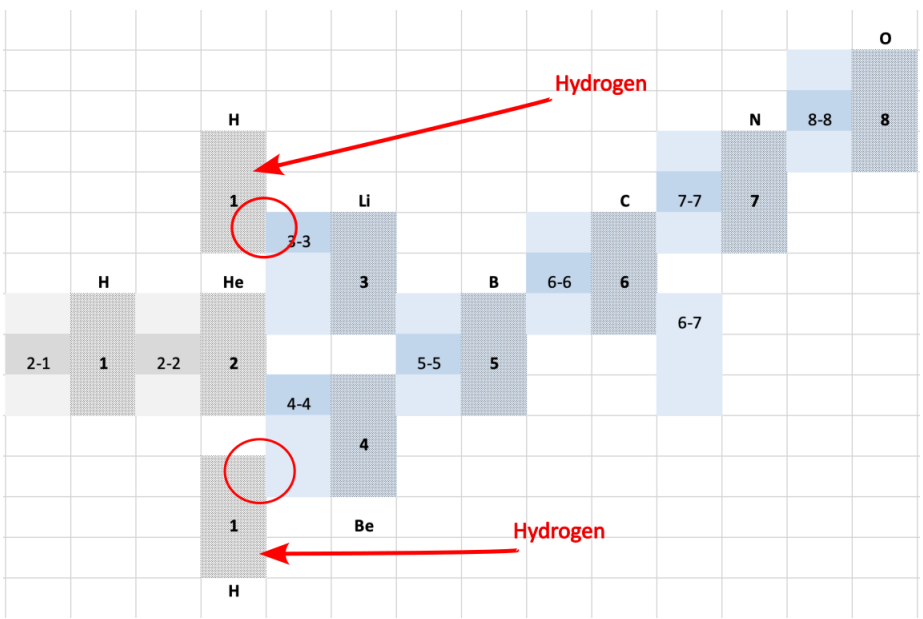


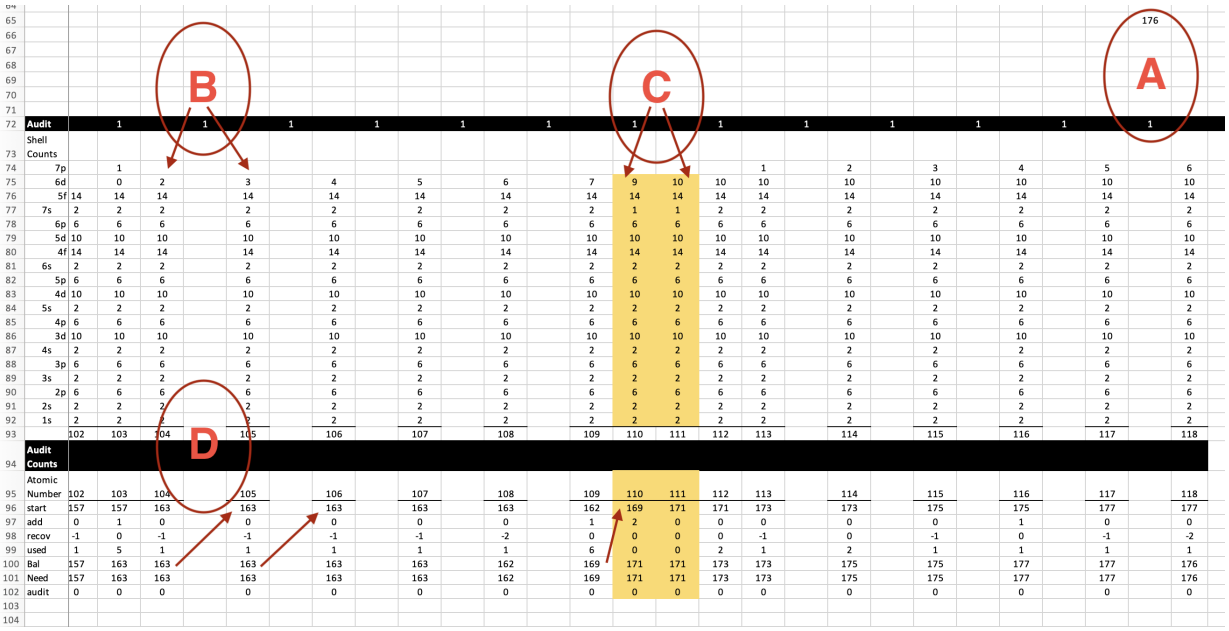
Figure 17: H_2O Molecule (Water)

3.4 Audits

The area below the Graphic build chart, beginning after spreadsheet row 71, contains an audit of the protons and neutrons used to illustrate the chart. Rows 72 through 93 tally the subshells for all shells used to document the elements. Rows 94 through 102 provide an audit tally for rows 72 through 93.

Audits are characterized by the following items:

- A: The count of neutrons on the chart. The number at the top of the oval labeled A, 176, is the horizontal tally of row 72, representing the total number of neutrons detailed in the chart.
- B: Oval B illustrates the tally of protons in the column above, with the total located in row 93.



65																			176
72	Audit	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
73	Shell Counts																		
74	7p	1										1	2	3	4	5	6		
75	6d	0	2	3	4	5	6	7	9	10	10	10	10	10	10	10	10		
76	5f	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14		
77	7s	2	2	2	2	2	2	2	1	1	2	2	2	2	2	2	2		
78	6p	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6		
79	5d	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10		
80	4f	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14		
81	6s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
82	5p	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6		
83	4d	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10		
84	5s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
85	4p	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6		
86	3d	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10		
87	4s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
88	3p	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6		
89	3s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
90	2p	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6		
91	2s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
92	1s	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
93		102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	
94	Audit Counts																		
95	Atomic Number	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	
96	start	157	157	163	163	163	163	163	162	169	171	171	173	173	175	175	177	177	
97	add	0	1	0	0	0	0	0	1	2	0	0	0	0	0	1	0	0	
98	recov	-1	0	-1	-1	-1	-1	-2	0	0	0	0	-1	0	-1	0	-1	-2	
99	used	1	5	1	1	1	1	1	6	0	0	2	1	2	1	1	1	1	
100	Bal	157	163	163	163	163	163	162	169	171	171	173	173	175	175	177	177	176	
101	Need	157	163	163	163	163	163	162	169	171	171	173	173	175	175	177	177	176	
102	audit	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Figure 18: Graphic Audits

- C: Columns are shown side by side, with colored shading and no blank columns between them. This indicates nonsequential protons listed in the columns above. For example, see columns FV and FW highlighted in the Graphic Audits figure. These columns do not have the proper protons listed in the build rows above the audit rows for elements numbered 110 and 111. These elements are located in a separate subshell and detailed in previous columns. Darmstadtium, element number 110, is found in column EM, and roentgenium, element number 111, is found in column EO. In these cases, as a visual aid, the out-of-order elements are highlighted in the shell's color coding without column spacing.

- D: Neutron audits. Begin with element number 104 and note that it has an ending neutron balance in row 100 of 163. That value carries forward as the beginning balance for element 105. Here, one neutron is removed and one is added, resulting in no net change. The same situation carries forward for element 106.

For element 109, one neutron was added in row 97, none were recovered, and six were used, resulting in a net addition of seven neutrons to the ending balance. That balance carries forward to element 110. Row 102 contains a formula that should always equal zero. Any other value indicates an error in processing the build.